Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/11/2005
PI Name:	Matula, Thomas J. Ph.D.		
Project Title:	Buoyancy-Driven Instabilities in Single-Bubble Sonoluminescence		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	matula@apl.washington.edu	**Fax:**	FY 206-543-6785
PI Organization Type:	UNIVERSITY	**Phone:**	206-685-7654
Organization Name:	University of Washington		
PI Address 1:	Applied Physics Lab		
PI Address 2:	1013 NE 40th St.		
PI Web Page:	http://www.apl.washington.edu/		
City:	Seattle	**State:**	WA
Zip Code:	98105-6698	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	12/01/2003	**End Date:**	10/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hall, Nancy Rabel	**Contact Phone:**	216-433-5643
Contact Email:	nancy.r.hall@nasa.gov		
Flight Program:			
Flight Assignment:	6/13/07--found out grant ended 10/1/2005 per N. Hall/GRC		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Crum, Lawrence (University of Washington)		
Grant/Contract No.:	NNC04GA51G		
Performance Goal No.:			
Performance Goal Text:			
	The principle objectives of this study are to determine how gravity affects the emission of light from single-bubble sonoluminescence (SBSL), and whether or not the bubble extinction is directly related to gravity. Our experimental task involves designing glass or quartz spherical levitation cells that generate very stable SL bubbles. The cells must have minimized vibration, and some temperature control. The experimental system will reside in a light-tight enclosure. Aside from acceleration, the frequency, pressure amplitude, and light intensity must be measured. A computer program will be constructed to perform all aspects of the experiment. The object of this research program is to quantify buoyancy-induced instabilities that may play a dominant role in the mechanism for sonoluminescence extinction. Such instabilities should be manifest in the long-term stability of the sonoluminescence bubble, and in the maximum forcing amplitude that can sustain stable sonoluminescence. Thus, light-emission data will be recorded under two conditions. (1)		

Task Description:	During constant forcing conditions, whereby the long-term stability of the light emission will be recorded, and (2) during a gradual approach to the extinction threshold. Furthermore, the bubble size (diameter) must also be measured, at its equilibrium (or natural) diameter, and again at its maximum diameter. These measurements will then be used in simulations of bubble dynamics to test current models for bubble extinction. The significance of this research goes far beyond the physics research laboratory. Determining how instabilities affect sonoluminescence will lead to understanding the mechanism for sonoluminescence. Understanding the role of instabilities in sonoluminescence, possible to the observance of hard x-ray emission, or even fusion, albeit over small temporal and spatial scales. The potential uses include a tabletop experimental device that can be used to test against computer simulations for implosions, such at those used in nuclear research centers. Furthermore, these studies should help in understanding the relatively new field of sonochemistry, which relies on cavitation bubbles for initiating or facilitating chemical reactions. Other uses not yet dreamed of, because of the current limitations, may become apparent if cavitation bubbles can concentrate more energy. At the very least, there exist the potential of exotic physics that can occur under the extreme conditions that sonoluminescence bubble experience, provided the energy concentration can be increased.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the dominant instabilities may help contribute to development of high-energy collapse dynamics in cavitation bubbles, such as the recently published experiment in cavitation fusion.
Task Progress:	The major objective of this year was to finish the transition of the engineering aspects from NASA engineers to the PI's lab. Major work was performed to reassemble and test the components. Additional tests were performed to evaluate specific technology generated for this work. In particular, a High-Speed Pixilated Spectrum Detector, designed and manufactured by Burle Industries, was tested. Significant accomplishments were achieved. The high-speed pixilated spectrum detector was able to observe band emissions from sonoluminescence bubbles during the transition phase. We had hypothesized that emission bands may be observable, assuming the transition to steady-state involved a particular pathway (a gradual heating of the bubble interior, rather than an abrupt transition). Observations of the emission bands are evidence for this hypothesized pathway. SUPPLEMENTAL INFORMATION FOR FINAL REPORT: The BDiSL project was downgraded from a flight status to a ground status project. This occurred after the project had already passed the SCR and RDR phases. Major engineering designs had been performed, and several bench-top prototype test beds were built. To take advantage of the designs and apparatus that was built, much of the equipment was sent to the PI's laboratory for continued work. APL received approximately 13 boxes, which included the major pieces of equipment, some supplies, and design drawings. The items were labeled prior to shipping, and inspected upon arrival. Further identification and labeling were done at the PI's lab. NASA property stickers were placed on all significant items. At least two audits of this equipment were performed over the year. All items targeted were correctly labeled and itemized, and had been properly documented in APL's inventory list. Additional policies have been implemented to track supplies and disposable items. Information about equipment and supplies is available from APL inventory personnel. The boxes were unpacked, and the experiment was reassembled. Major reassembly components included the SBSL system, the imaging system, the spectral system, and the computer control system. A Master's level graduate student was employed to reassemble the system, and to make it operational. The student verified that the SBSL system was working properly, using several gas and water mixtures. Computer control of the experiment was verified. The spectral system was also assembled and tested successfully. The important technology developed under this grant – the pixilated high speed spectrum detector (PHSSD) – was subjected to various in-house tests to determine its speed and resolution for SBSL bubbles. The PI and student performed a series of experiments to test the complete system. The idea was to measure SBSL spectra in the steady-state, during the transition to sonoluminescence, and at the extinction threshold. Steady-state spectra are a good starting point for testing the system because numerous data sets are available for comparison. The results are typical of what is observed in SBSL. The spectrum increases as the wavelength decreases. There is no peak in the spectrum. Typically, one uses blackbody fitting to obtain a 'temperature' of the bubble interior. An experiment was designed to test the acquisition speed of the High Speed Pixilated Spectrum Detector: During the startup phase, the bubble undergoes a transition from a non-sonoluminescing to a sonoluminescing state. Previous studies by us suggested that the time frame was very fast, but the light intensity increased gradually. Because of this gradual increase in light emission, we hypothesized that during the initial stages of transition (which we call the incipient luminescent state), we may be able to observe band emission from the bubble. This is significant because currently, SBSL does not show any band emission. Indeed, the only observation of band emissions from single bubbles were performed with unstable steady-state bubbles. Our goal was to determine whether or not band spectra could be measured during the incipient phase of SBSL, as the bubble transitioned into a steady state. During the start-up phase, the spectral system collected spectra for up to one minute. The spectra were compared to a spectrum of a stable sonoluminescing bubble. Remarkably, the experiments showed that indeed, there is a short-lived emission band near 310 nm that becomes obscured by the continuum as the light intensity increases. This remarkable discovery – and verification of our hypothesis – proves that the bubble gets hotter as it transitions towards steady-state; the emission bands becoming obscured as the bubble heats up. The results of this experiment were presented at an Acoustical Society of America conference during a special session titled "Sonoluminescence, sonochemistry, and sonofusion". A manuscript for publication is being planned.
Bibliography Type:	Description: (Last Updated: 02/27/2004)